

teiphy: A Python Package for Converting TEI XML Collations to NEXUS and Other Formats

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Summary

Textual scholars have been using phylogenetics to analyze manuscript traditions since the early 1990s (Robinson & O'Hara, 1992). Many standard phylogenetic software packages accept as input the NEXUS file format (Maddison et al., 1997). The teiphy program takes a collation of texts encoded in TEI XML format and can convert it to any of the following formats amenable to phylogenetic analysis: NEXUS (with support for ambiguous states and clock model calibration data blocks for MrBayes or BEAST2), Hennig86, PHYLIP (relaxed for use with RAxML), FASTA (relaxed for use with RAxML), and STEMMA (designed for Stephen C. Carlson's stemmatic software tailored for textual data). For machine learning-based analyses, teiphy can also convert a TEI XML collation to a collation matrix in NumPy, Pandas DataFrame, CSV, TSV, or Excel format.

Statement of Need

For over a decade, the Text Encoding Initiative has endeavored to provide an international standard for digitally encoding textual information for the humanities (Ide & Sperberg-McQueen, 1995). Their guidelines describe a standard format for encoding material details, textual transcriptions, and critical apparatuses (TEI Consortium, 2022). Due to its rich and well-documented set of elements for expressing a wide range of features in these settings, the Text Encoding Initiative's Extensible Markup Language format (hereafter abbreviated TEI XML) has become the *de facto* format for textual data in the digital humanities (Fischer, 2020). Its expressive power has proven increasingly valuable since its release, as scholars have learned—sometimes the hard way—that digital transcriptions and collations should

1. preserve as much detail as they can from their material sources, including paratextual features;
2. reproduce the text of their sources as closely as possible, with editorial regularizations to things like orthography, accentuation, and scribal shorthand encoded alongside rather than in place of the source text; and
3. describe uncertainties about a source's contents as accurately as possible, allowing for degrees of uncertainty and multiple choices for disambiguations if necessary.

These principles have much bearing on the editing of critical texts, a task fundamental to the field of philology. Phylogenetic algorithms developed in the context of evolutionary biology have been popular approaches to this task, especially as philology itself has taken a digital turn over the years. Taking the most arduous part of reconstructing a textual tradition and delegating it to a computer proved to be a promising technique, and its successful demonstration with a portion of *The Canterbury Tales* was a milestone in the development of the field (Barbrook et al., 1998). Soon after this, the same methods were applied more comprehensively to the tradition of *Lanseloet van Denemerken* in a work that would formalize many practical rules

for computer-assisted textual criticism (Salemans, 2000). Over the decades preceding and following these developments, biologists have continued to develop and improve phylogenetic methods (Felsenstein, 2004), and textual critics have adapted these improvements and even added their own innovations to make the process more suitable for their purposes (Carlson, 2015; Edmondson, 2019; Hyytiäinen, 2021; Spencer et al., 2002, 2004; Turnbull, 2020).

Phylogenetic algorithms have a natural place in textual criticism given the deep analogy between textual traditions and evolutionary trees of life: a sequence alignment, which consists of taxa, sites or characters, and the states of taxa at those characters, corresponds almost identically to a collation, which consists of witnesses to the text, locations of textual variation (which we will call “variation units” from here on), and the variant readings attested by witnesses at those points.

The problem is that no currently available phylogenetic software accepts inputs in TEI XML format. Instead, most phylogenetic programs expect inputs in NEXUS format (Maddison et al., 1997). This format was conceived with versatility—including use in stemmatic and non-biological applications—in mind, and this design choice has been vindicated in its general applicability, but NEXUS is neither equipped nor meant to express the same kinds of details that TEI XML is. Conversely, for those interested primarily in working with the collation as an alignment, TEI XML is overkill. Thus, a chasm continues to separate data born in the digital humanities from phylogenetic tools born in the biological sciences, and the only way to bridge it is by conversion.

The challenge is made more daunting by the variety of tools available for phylogenetic and other analyses, some of which expect inputs other than NEXUS files or NEXUS files augmented in different ways. For instance, the cladistic software PAUP*, which has historically been the tool of choice for text-critical applications that evaluate candidate trees using the criterion of maximum parsimony, reads inputs in NEXUS format (Swofford, 2003), but the TNT software, its main competitor among maximum parsimony-based programs, expects inputs in Hennig86 format (Farris, 1988; Goloboff & Catalano, 2016). While Hennig86 format does not allow for as much flexibility in the input as NEXUS does (e.g., it does not support ambiguous character states that can be disambiguated as some states but not others), TNT’s extensive support for morphological state models makes it potentially more suitable for textual data, and textual critics may prefer it to PAUP*. In the same regime, Stephen C. Carlson’s *STEMMA software*, initially developed for the cladistic analysis of biblical texts known to be affected by contamination, makes substantial adaptations to the basic maximum-parsimony phylogenetic approach to account for this problem and other constraints common in a text-critical setting (Carlson, 2015); however, the input collation data must be provided in a *STEMMA*-specific format. Likewise, among programs that use the maximum likelihood criterion instead of maximum parsimony, IQ-TREE accepts inputs in NEXUS format (Minh et al., 2020), but RAXML interfaces primarily with inputs in PHYLIP and FASTA format (Stamatakis, 2014). Finally, for phylogenetic programs that attempt to estimate the posterior distribution of candidate trees in a Bayesian fashion, MrBayes and BEAST2 both accept inputs in NEXUS format (or can convert NEXUS inputs to their standard input format), but they expect taxon dates (for the calibration of evolutionary clock models) to be specified in the NEXUS file in different code blocks (Bouckaert et al., 2019; Ronquist et al., 2012).

Furthermore, end users of textual collations may be interested in non-phylogenetic analyses. In this case, the desired input format is often not a NEXUS-style sequence alignment, but a collation matrix with a row for each variant reading and a column for each witness (or vice-versa). For Python machine-learning libraries like Scikit-learn (Pedregosa et al., 2011), NIMFA (Zitnik & Zupan, 2012), and TensorFlow (Abadi et al., 2015), the standard input format is a NumPy array (Harris et al., 2020), although Pandas DataFrames, which support row and column labels (McKinney, 2010; The pandas development team, 2020), may also be supported. (The latter format also extends the conversion pipeline to many other formats, including CSV, TSV, and Excel files; Pandas DataFrames can even write their contents to database tables.) To give an example, the text of the New Testament has served as a testbed

for multiple analyses of this type, which have generally applied clustering and biclustering algorithms to collation matrices (Baldwin, 2010; Finney, 2018; McCollum, 2019; Thorpe, 2002; Willker, 2008). Given the prevalence of efforts like these, the need for a means of converting TEI XML collations to NumPy collation matrices or labeled Pandas DataFrames is clear.

Design

While the conversion process is a straightforward one for most collation data, various sources of ambiguity can make a one-to-one mapping of witnesses to readings impossible. One such source of ambiguity is lacunae, or gaps in the text due to erasure, faded ink, or damage to the page. Another is retroversions, or readings in the original language of the text reconstructed through the back-translation of subsequent versions of the text in other languages. Mechanisms for modeling ambiguous states resulting from situations like these exist in both TEI XML and NEXUS, and in both parsimony- and likelihood-based phylogenetic methods, ambiguities about the states at the leaves and even at the root of the tree can be encoded and leveraged in the inference process. For these reasons, it is imperative to ensure that these types of judgments, as well as other rich features from TEI XML, can be respected (and, where necessary, preserved) in the conversion process.

Collations should preserve as much detail as possible, including information on how certain types of data can be normalized and collapsed for analysis. Since one might want to conduct the same analysis at different levels of granularity, the underlying collation data should be available for use in any case, and only the output of the conversion should reflect changes in the desired level of detail. Likewise, as noted in the previous section, uncertainty about witnesses' attestations should be encoded in the collation and preserved in the conversion of the collation.

For text-critical purposes, differences in granularity typically concern which types of variant readings we consider important for analysis. At the lowest level, readings with uncertain or reconstructed portions are almost always considered identical with their reconstructions (provided these reconstructions can be made unambiguously) for the purpose of analysis. Defective forms that are obvious misspellings of a more substantive reading are often treated the same way. Even orthographic subvariants that reflect equally "correct" regional spelling practices may be considered too common and of too trivial a nature to be of value for analysis. Other readings that do not fall under these rubrics but are nevertheless considered manifestly secondary (due to late and/or isolated attestation, for instance), may also be considered uninformative "noise" that is better left filtered out.

Use Case

Due to the availability of extensive collation data for the Greek New Testament, and because this project was originally developed for use with such data, we tested this library on a sample collation of the book of Ephesians in thirty-eight textual witnesses (including the first-hand texts of manuscripts, corrections made to manuscripts by later hands, translations to other languages, and quotations from church fathers). The manuscript transcriptions used for this collation were those produced by the University of Birmingham's Institute for Textual Scholarship and Electronic Editing (ITSEE) for the International Greek New Testament Project (IGNTP); they are freely accessible at <https://itseeweb.cal.bham.ac.uk/epistulae/XML/igntp.xml>. To achieve a balance between variety and conciseness, we restricted the collation to a set of forty-two variation units in Ephesians corresponding to variation units in the United Bible Societies Greek New Testament (Aland et al., 2014), which highlights variation units that affect substantive matters of translation.

In our example collation, witnesses are described in the `listWit` element under the `teiHeader`. Because most New Testament witnesses are identified by numerical Gregory-Aland identifiers,

these witnesses are identified with `@n` attributes; the recommended practice is to identify such elements by `@xml:id` attributes, but this software is designed to work with either identifying attribute (preferring `@xml:id` if both are provided), and we have left things as they are to demonstrate this feature.

The witness elements in the example collation also contain `origDate` elements that provide dates or date ranges for the corresponding witnesses. Where a witness can be dated to a specific year, the `@when` attribute is sufficient to specify this; if it can be dated within a range of years, the `@from` and `@to` attributes or the `@notBefore` and `@notAfter` attributes should be used; the software will work with any of these options. While such dating elements are not required, our software includes them in the conversion process whenever possible. This way, phylogenetic methods that employ clock models and other chronological constraints can benefit from this information when it is provided.

Each variation unit is encoded as an `app` element with a unique `@xml:id` attribute. Within a variation unit, a `lem` element without a `@wit` attribute presents the main text, and it is followed by `rdg` elements that describe variant readings (with the first `rdg` duplicating the `lem` reading and detailing its witnesses) and their attestations among the witnesses. (Situations where the `lem` reading is not duplicated by the first `rdg` element, but has its own `@wit` attribute, are also supported.) For conciseness, we use the `@n` attribute for each reading as a local identifier; the recommended practice for readings that will be referenced elsewhere is to use the `@xml:id` attribute, and this software will use this as the identifier if it is specified, but we have only specified `@xml:id` attributes for `rdg` elements referenced in other variation units to demonstrate the flexibility of the software. For witnesses with missing or ambiguous readings at a given variation unit, we use the `witDetail` element. For ambiguous readings, we specify their possible disambiguations with the `@target` attribute and express our degrees of certainty about these disambiguations using `certainty` elements under the `witDetail` element.

The [TEI XML file](#) for this example is available in the example directory of the GitHub repository. Full instructions for converting this file using `teiphy` and analyzing it with several different phylogenetic packages are provided in the documentation, but here, we will walk through the command-line arguments involved in installing `teiphy` and using it to convert our example TEI XML collation (1) to a NEXUS file suitable for use with IQ-TREE, and (2) to input for the STEMMA program.

Because `teiphy` is published in the [Python Package Index \(PyPI\)](#), it can be installed via the command

```
pip install teiphy
```

Now we are ready to convert our TEI XML collation to a NEXUS file for IQ-TREE. Let us suppose that we would like to treat reconstructions of unclear or missing text, defective spellings, and orthographic variations in spelling as trivial variants for the purposes of our phylogenetic analysis. We can specify this to `teiphy` with the `-t` flag for each trivial type of reading as follows:

```
-t reconstructed -t defective -t orthographic
```

In addition, suppose we would like to treat placeholders for overlapping variants from larger units and lacunae as missing data. We can specify this to `teiphy` with the `-m` flag for each type of reading to be read as missing data as follows:

```
-m overlap -m lac
```

If, in variation units where manuscripts are corrected or have alternate readings provided by other hands, our collation adds `*` and `T` suffixes to manuscript sigla to mark the work of the original hand, we can tell `teiphy` to ignore these suffixes using the `-s` flag with each trivial suffix:

```
-s "*" -s T
```

192 (Note that because the * character is reserved on the command-line, we must place it between
193 quotation marks directly after the -s flag.)

194 When corrections are made to a manuscript, they are typically sporadic, and as a result, the
195 text of corrector witnesses like 06C1 and 06C2 will tend to be too fragmentary to be useful for
196 analysis. But if we wish to assume that each corrector tacitly adopted all of the readings from
197 the previous hand that he or she did not change, then teiphy can “fill out” each corrector’s
198 text using the text of the first hand (in the case of the first corrector) or the filled-out text of
199 the previous corrector (for all subsequent correctors). Thus, 06C1 would replicate the text of
200 06* (i.e., the first hand responsible for the text of 06) where it does not introduce its own
201 readings, and 06C2 would then replicate the text of 06C1 where it does not introduce its own
202 readings. If we want to apply this transformation during the conversion process, then we can
203 specify this with the --fill-correctors flag.

204 Finally, we must specify the required arguments to teiphy, which are the input TEI XML
205 file (example/ubs_ephesians.xml) and the name of the output NEXUS file (ubs_ephesians-
206 iqtree.nexus). Note that we do not have to specify the desired output format explicitly;
207 teiphy will determine from the output filename that it should write a NEXUS file. Combining
208 the previous options and arguments, the complete command is

```
209 teiphy -t reconstructed -t defective -t orthographic -m overlap -m lac  
210 -s "*" -s T --fill-correctors  
211 example/ubs_ephesians.xml ubs_ephesians-iqtree.nexus
```

212 If we pass the resulting NEXUS file to IQ-TREE and specify appropriate settings for our textual
213 data (in this case, the Lewis Mk substitution model with ascertainment bias correction), we
214 will get an output tree like the one shown in Figure 1.

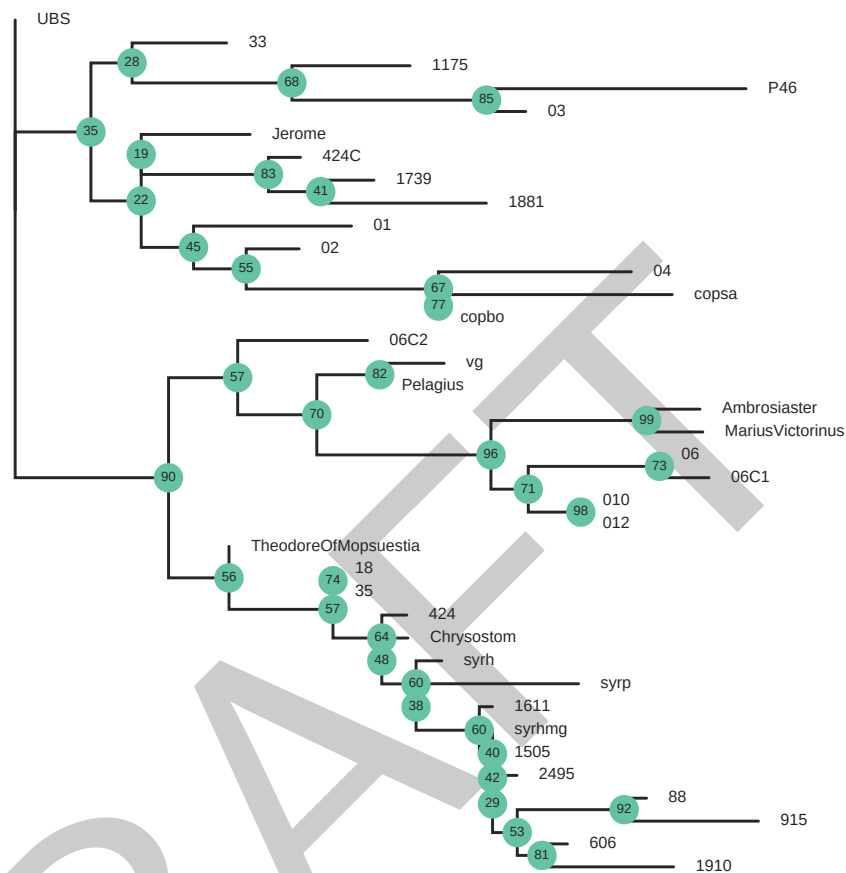


Figure 1: A phylogenetic tree inferred by IQ-TREE for the UBS Ephesians example data with support values on the branches based on 1000 bootstrap replicates. Reconstructed, defective, and orthographic sub-variants were treated as identical to their parent readings, and changes made to the text by later correctors (represented as distinct witnesses with sigla like 06C1 and 06C2) were filled in with the readings of the first hand or the previous corrector where the corrector was not active.

215 If we want to generate input files for the STEMMA program using the same options, only a few
216 adjustments are required. First, since multiple files are written for STEMMA input (namely,
217 a collation file with no file extension and a .chron file containing information about witness
218 dates), we only specify the base of the filename for our output. Second, since the filename
219 now has no extension, we must specify the desired output format to teiphy with the argument

220 `--format stemma`

221 Combining the options and arguments we used before with these changes, the complete
222 command is

```
223 teiphy -t reconstructed -t defective -t orthographic -m overlap -m lac
224 -s "*" -s T --fill-correctors --format stemma
225 example/ubs_ephesians.xml stemma_example
```

226 If we process the output files with the PREP utility that accompanies STEMMA and then pass
227 the resulting files to STEMMA, we will get an output tree like the one shown in Figure 2.

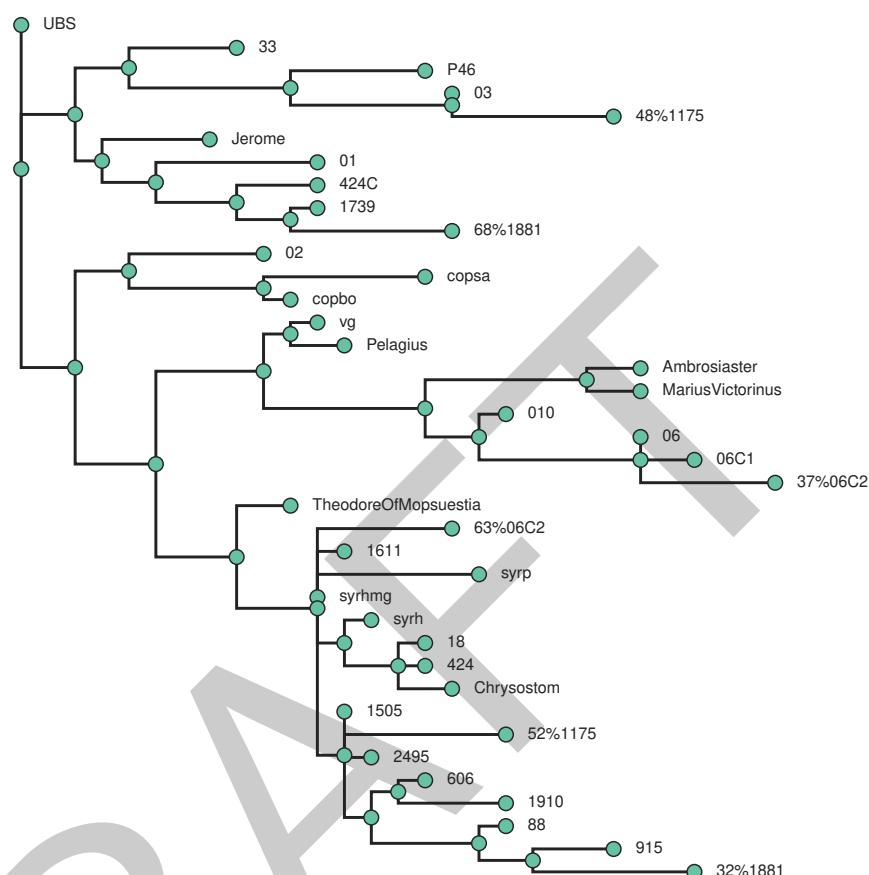


Figure 2: A phylogenetic tree inferred by STEMMA for the UBS Ephesians example data using 100 iterations of simulated annealing. Mixed witnesses are split (with proportions of their readings indicated by the percentages before their sigla) and located at different parts of the tree. Note that some witnesses (e.g., 012, 35) from the collation are excluded from this tree by STEMMA because they have the same reading sequence as another witness after their reconstructed, defective, and orthographic readings have been regularized.

For the small sample of variation units covered in the UBS apparatus for Ephesians, the phylogenetic results depicted in Figures 1 and 2 are impressive. The trees produced by IQ-TREE and STEMMA agree on several traditionally established groupings of manuscripts, including Family 1739 (1739, 1881, and the corrections to 424); the “Western” tradition (as preserved in the Greek-Latin diglots 06, 010, and 012, the Latin Vulgate, and the early Latin church fathers Ambrosiaster, Marius Victorinus, and Pelagius); and the later Byzantine tradition (with representative manuscripts 18 and 35 and church fathers Chrysostom and Theodore of Mopsuestia). The Harklean Syriac translation (syrh) and the witnesses to its Greek *Vorlage* (1505, 1611, 2495) are correctly placed within the Byzantine tradition, although the two programs disagree on how to describe their relationships within that tradition. While IQ-TREE does not account for mixture complicating the tradition, STEMMA identifies three witnesses suspected to exhibit Byzantine contamination: 1175, 1881, and the second corrector of 06. Both programs also identify the Codex Alexandrinus (02) as closely related to both the Sahidic and Bohairic Coptic translations of Ephesians (copsa, copbo), although they disagree on where this clade is located in the larger tradition. Despite their discrepancies regarding

certain subtrees, the extent of their agreements speaks to the level of genealogically significant detail preserved in the TEI XML apparatus and the NEXUS and STEMMA inputs generated from it.

Availability

As noted above, the software is published in [PyPI](#) and can be installed from there using `pip`. The source code is available under the MIT license from the [GitHub repository](#). The automated testing suite has 100% coverage, and functional tests where our example TEI XML file is converted and run through RAXML, IQ-TREE, MrBayes, and STEMMA are part of teiphy's continuous integration (CI) pipeline.

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